

Chapter 3

A is invertible



$\det(A) \neq 0$

Properties of Determinants

- Basic Property 1: $\det(I) = 1$
- Basic Property 2: Exchange rows reverse the sign of det
 - If a matrix A has 2 equal rows, $\det A = 0$
- Basic Property 3: Determinant is “linear” for each row

$$\det \begin{pmatrix} ta & tb \\ c & d \end{pmatrix} = t \det \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\det \begin{pmatrix} a + a' & b + b' \\ c & d \end{pmatrix} = \det \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \det \begin{pmatrix} a' & b' \\ c & d \end{pmatrix}$$

- A row of zeros, $\det A = 0$
- “Subtract k x row i from row j ” does not change det

Determinants for Upper Triangular Matrix

$$U = \begin{bmatrix} d_1 & \cdots & * \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{bmatrix}$$

Killing everything above
Does not change the det

$$\det(U) = \det \left(\begin{bmatrix} d_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & d_n \end{bmatrix} \right)$$

3-a Property 1

$$= d_1 d_2 \cdots d_n \det \left(\begin{bmatrix} 1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & 1 \end{bmatrix} \right)$$

= 1

$$\det(U) = d_1 d_2 \cdots d_n \text{ (Products of diagonal)}$$

More Properties of Determinants

- $\det(AB) = \det(A)\det(B)$

$$\det(A + B) \neq \det(A) + \det(B)$$

Q: find $\det(A^{-1})$

$$\because A^{-1}A = I \quad \therefore \det(A^{-1})\det(A) = \det(I) = 1$$

$$\therefore \det(A^{-1}) = 1/\det(A)$$

Q: find $\det(A^2)$

$$\det(A^2) = \det(A)\det(A) = \det(A)^2$$

- $\det(A^T) = \det(A)$

- Zero row $\rightarrow$ zero column
- Same row $\rightarrow$ same column

3-1 80. Calculate the determinant of $\begin{bmatrix} O & I_m \\ I_n & O \end{bmatrix}$

3-2 71. Let A and B be $n \times n$ matrices such that B is invertible. Prove that $\det(B^{-1}AB) = \det A$.

$$\det B^{-1} = 1/\det B$$

$$\det(B^{-1}AB) = \det B^{-1} \det A \det B$$

$$= \frac{1}{\cancel{\det B}} \det A \cancel{\det B}$$

$$= \det A$$

Review 42. Suppose that an $n \times n$ matrix can be expressed in the form $A = PDP^{-1}$, where P is an invertible matrix and D is a diagonal matrix. Prove that the determinant of A equals the product of the diagonal entries of D .

$$\begin{aligned} \det A &= \det (PDP^{-1}) = \det D \\ &= \det \begin{pmatrix} d_1 & & 0 \\ & d_2 & \\ 0 & & \ddots \\ & & & d_n \end{pmatrix} = d_1 d_2 \cdots d_n \end{aligned}$$

3-2 72. An $n \times n$ matrix A is called nilpotent, if for some positive integer k , $A^k = 0$. Prove that if A is nilpotent, then $\det A = 0$.

$$\det A^k = (\det A)^k = \det 0 = 0$$

$$\Rightarrow \det A = 0$$

3-2 73. An $n \times n$ matrix Q is called orthogonal if $Q^T Q = I_n$. Prove that if Q is orthogonal, then $\det Q = \pm 1$.

$$\det(Q^T Q) = \det I_n = 1$$

$$\begin{aligned} \Rightarrow \det(Q^T Q) &= \det(Q^T) \det Q \\ &= (\det Q)^2 = 1 \end{aligned}$$

$$\Rightarrow \det Q = \pm 1$$

3-2 74. A square matrix A is called skew-symmetric if $A^T = -A$. Prove that if A is a skew-symmetric $n \times n$ matrix and n is odd, then A is not invertible. What if n is even?

$$\begin{aligned}
 A^T &= -A && n \text{ is odd} \\
 \Rightarrow \det A^T &= \det(-A) && ??? \\
 \Rightarrow \det A &= -\det A \\
 \Rightarrow \det A &= 0 \\
 \Rightarrow A &\text{ is not inv.}
 \end{aligned}$$

$$\begin{aligned}
 \det(kA) &\neq k \det A \\
 k \left(\begin{array}{c} \equiv \\ \equiv \\ \equiv \end{array} \right) &\parallel k^n \det A \\
 A: n \times n
 \end{aligned}$$

$$\begin{aligned}
 \det A^T &= \det(-A) \\
 \parallel \det A &= (-1)^n \det A \\
 n \text{ odd: } \det A &= -\det A = 0 && \text{not inv.} \\
 n \text{ even: } \det A &= \det A && \text{Inv? } \checkmark \\
 &&& \text{not inv? } \checkmark
 \end{aligned}$$